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21.5 Spheres and Rotations

The surface of the Earth is called a 2-sphere by topologists, but a 3-sphere by geometers; so the term n -sphere is somewhat unclear. We'll say "sphere in n dimensions" to avoid any ambiguity. (For Earth, $n = 3$.) *Sphere* refers to the surface, *ball* to the interior volume.

A sphere of radius r in n dimensions, centered on the origin, is the locus of points for which

$$x_0^2 + \cdots + x_{n-1}^2 = r^2 \quad (21.5.1)$$

Points on the sphere in n dimensions can be specified by $n - 1$ angular coordinates, roughly the analogs of latitude and longitude on the sphere in three dimensions,

$$\begin{aligned} x_0 &= r \cos \psi_0 \\ x_1 &= r \sin \psi_0 \cos \psi_1 \\ &\dots \\ x_{n-2} &= r \sin \psi_0 \sin \psi_1 \cdots \cos \psi_{n-2} \\ x_{n-1} &= r \sin \psi_0 \sin \psi_1 \cdots \sin \psi_{n-2} \end{aligned} \quad (21.5.2)$$

All the angles except the last have the range

$$0 \leq \psi_i \leq \pi, \quad i = 0, \dots, n-3 \quad (21.5.3)$$

i.e., are "latitude-like." The last angle is "longitude-like,"

$$0 \leq \psi_{n-2} \leq 2\pi \quad (21.5.4)$$

The surface area S_n of the sphere in n dimensions has a simple recurrence,

$$\begin{aligned} S_1 &= 2 && \text{(two points)} \\ S_2 &= 2\pi r && \text{(circumference of circle)} \\ S_n &= \frac{2\pi r^2}{n-2} S_{n-2}, && n > 2 \end{aligned} \quad (21.5.5)$$

The volume V_n of the n -dimensional ball is equal to r/n times the area of the enclosing sphere in n dimensions, and also has a simple recurrence,

$$\begin{aligned} V_1 &= 2r && \text{(length of line)} \\ V_2 &= \pi r^2 && \text{(area of circle)} \\ &\dots && \\ V_n &= \frac{r}{n} S_n = \frac{2\pi r^2}{n} V_{n-2} \end{aligned} \quad (21.5.6)$$

Closed-form formulas require a gamma function,

$$\begin{aligned} S_n &= \frac{2\pi^{n/2}}{\Gamma(\frac{1}{2}n)} r^{n-1} \\ V_n &= \frac{2\pi^{n/2}}{n\Gamma(\frac{1}{2}n)} r^n \end{aligned} \quad (21.5.7)$$

As n becomes large, the ratio of the volume of a ball to the volume of the circumscribed (hyper-) cube rapidly becomes small,

$$\frac{V_n}{2^n} \rightarrow 0, \quad n \rightarrow \infty \quad (21.5.8)$$

21.5.1 Picking a Random Point on the Sphere

You don't get a random point on the sphere in n dimensions by picking uniformly random values for the $n - 1$ angles in equation (21.5.2), just as you don't get a random point on the Earth's surface by throwing darts at a Mercator map (or any other non-equal-area projection).

An elegant general method is to generate n independent, identically distributed, normal (Gaussian) deviates of zero mean, say $y_0, \dots, y_{n-1}$ (see §7.3), and then calculate a point $\mathbf{x}$ on the unit sphere in n dimensions by

$$\mathbf{x} = \frac{\mathbf{y}}{|\mathbf{y}|} \quad (21.5.9)$$

or, in other words,

$$x_i = y_i / \sqrt{\sum_{j=0}^{n-1} y_j^2} \quad (21.5.10)$$

This works because the spherically symmetric Gaussian distribution in n dimensions trivially factorizes into a product of independent one-dimensional Gaussians. If you want a random point inside the enclosed n -volume, generate an additional *uniform* random deviate u in $[0, 1]$ and calculate the point's coordinates as

$$x_i = u^{1/n} y_i / \sqrt{\sum_{j=0}^{n-1} y_j^2} \quad (21.5.11)$$

You can of course scale to any other radius of sphere.

Faster special methods are available for the spheres in two, three, and four dimensions. For two dimensions, the circle, pick u_0 and u_1 uniform in $[-1, 1]$, rejecting choices for which $u_0^2 + u_1^2 > 1$. This picks a random point inside the unit circle. Now scale in the obvious way to get a point on the circle,

$$x_0 = \frac{u_0}{\sqrt{u_0^2 + u_1^2}}, \quad x_1 = \frac{u_1}{\sqrt{u_0^2 + u_1^2}} \quad (21.5.12)$$

(We already discussed this method in §7.3, under Cauchy deviates.)

A faster method for three dimensions, also using only two random deviates, is due to Marsaglia [1]. Pick a point *inside* the unit circle (u_0, u_1) as above. Then a random point on the sphere in three dimensions is

$$\begin{aligned} x_0 &= 2u_0 \sqrt{1 - u_0^2 - u_1^2} \\ x_1 &= 2u_1 \sqrt{1 - u_0^2 - u_1^2} \\ x_2 &= 1 - 2(u_0^2 + u_1^2) \end{aligned} \quad (21.5.13)$$

For the sphere in four dimensions, pick two independent points *inside* the unit circle as above, (u_0, u_1) and (u_2, u_3) . Then a random point on the sphere in four dimensions is [1]

$$\begin{aligned} x_0 &= u_0 \\ x_1 &= u_1 \\ x_2 &= u_2 \sqrt{\frac{1 - u_0^2 - u_1^2}{u_2^2 + u_3^2}} \\ x_3 &= u_3 \sqrt{\frac{1 - u_0^2 - u_1^2}{u_2^2 + u_3^2}} \end{aligned} \quad (21.5.14)$$

Unfortunately, there is no known generalization to higher dimensions.

21.5.2 Picking a Random Rotation Matrix

Don't confuse this with picking a point on a sphere. A rotation matrix $\mathbf{M}$ in n dimensions is an orthogonal $n \times n$ matrix. For a *proper rotation*, $\mathbf{M}$ must have determinant 1. The other possibility, determinant -1 , represents an *improper rotation*, decomposable into a proper rotation followed by a reflection. The rotation matrix $\mathbf{M}$ maps any point $\mathbf{x}$ to a new point $\mathbf{x}'$ by

$$\mathbf{x}' = \mathbf{M} \cdot \mathbf{x} \quad (21.5.15)$$

A general method for picking a uniformly random rotation matrix is to fill an $n \times n$ matrix $\mathbf{G}$ with independent, identically distributed, normal (Gaussian) deviates of zero mean. Then, use QRdcmp in §2.10 to construct the *QR* decomposition, namely $\mathbf{G} = \mathbf{Q} \cdot \mathbf{R}$. Except for the possibility that it might have the wrong sign of determinant, the matrix $\mathbf{Q}$ is now a uniformly random rotation matrix. The method used in QRdcmp is to apply $n - 1$ Householder transformations, each of which is